

HK IMO Training 2020-2021
Problem Set 7
For 5 December 2020



Problem 25. In a mathematical contest, there are 3 problems. A contestant can get an integer score between 0 and 7 (including 0 and 7) in each problem. It is known that for any two contestants, there exists at most one problem in which they have obtained the same score. For example, there are no two contestants whose scores are 6, 7, 0 and 6, 7, 1 in this order, but there might be two contestants whose scores are 6, 7, 0 and 7, 6, 0 in this order.

Find the maximum number of contestants.

Problem 26. There is a finite number of stones. The weight of each stone is not greater than 0.5kg, and the total weight of the stones is not greater than 2.5kg. Prove that the stones can be divided into 4 groups, so that each group has a total weight not greater than 1kg.

Problem 27. Let A_n be the set of all partitions of the sequence $1, 2, \dots, n$ into subsequences such that every two adjacent terms of each subsequence have different parities. Let B_n be the set of all partitions of the sequence $1, 2, \dots, n$ into subsequences such that all the terms of each subsequence have the same parity. For example, the partition $(1, 4, 5, 8), (2, 3), (6, 9), (7)$ is an element of A_9 , and the partition $(1, 3, 5), (2, 4), (6)$ is an element of B_6 .

Prove that for every positive integer n , the sets A_n and B_{n+1} contain the same number of elements.

Problem 28. Let p be an odd prime. There are n points on the plane with no three collinear. Any pair of points is joined by a segment. Each segment is coloured in one of the $p + 1$ given colours so that each colour is used at least once and the following condition holds: if two sides of a triangle have the same colour, then the third side also has the same colour.

Find the maximum value of n in terms of p for which such a colouring is possible.

(It would be the best if you have a solution with the odd prime p replaced by an arbitrary integer $m \geq 3$.)